Credit Risk Segmentation Analysis

Project Objective & Scope

Objective: Identify key borrower attributes separating default risk.

Dataset Scope: ~2.26M total loan records, using 36 selected features out of 145 columns.

Core Methodology: Automated driver ranking via Observed Default Rate (ODR) variation ranges.

Technical Data Engineering Pipeline

Data Ingestion: Loading 2.26M+ LendingClub loan records into Snowflake.

Cleaning & Filtering: Filtering closed loans and handling missing values.

Feature Engineering: Applying decile and bucket logic to calculate Observed Default Rate (ODR).

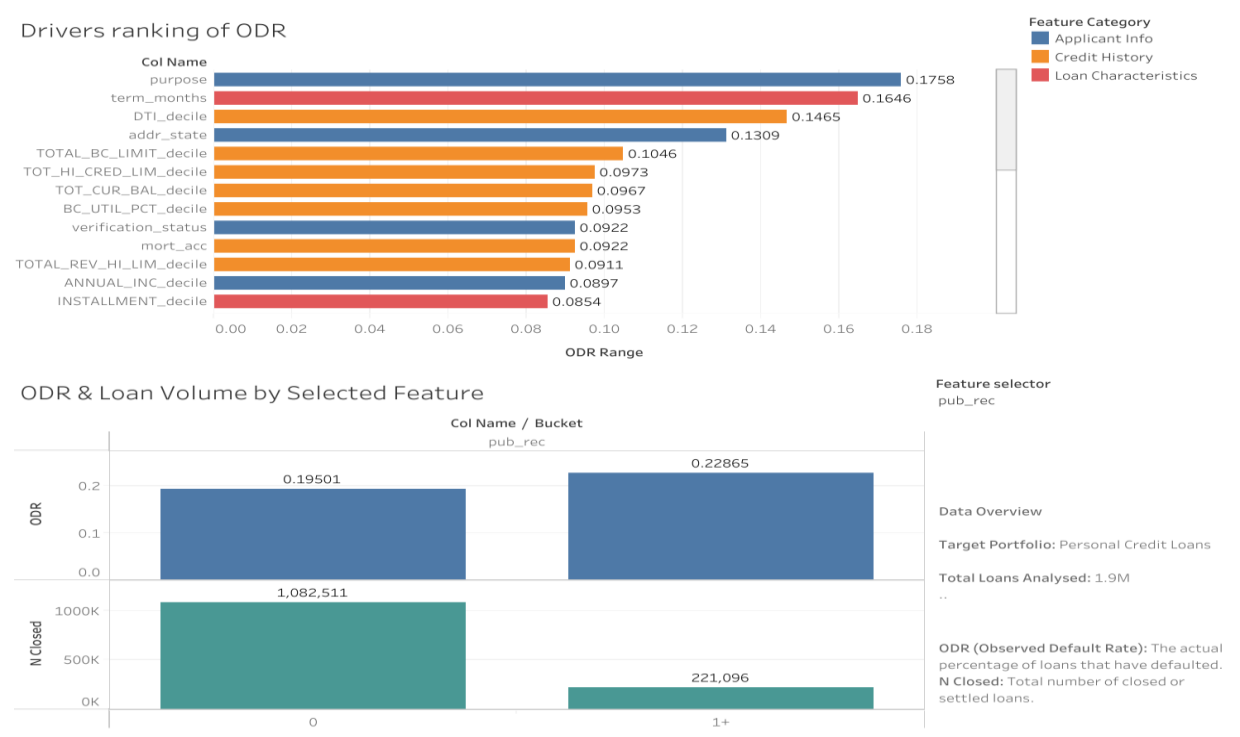
Driver Ranking Metric: Identifying high-impact risk drivers using ODR range metrics.

Key Risk Drivers & Dashboard

Key Risk Findings & Policy Implications

Instead of relying on a single ranked list, drivers were grouped to highlight structural portfolio vulnerabilities and their policy implications.

- **Loan Characteristics**: Loans with 60-month terms show consistently higher default risk than 36-month loans, suggesting longer maturities may require tighter approval criteria or risk-based pricing.
- **Credit History**: Borrowers with high debt-to-income (DTI) ratios show a monotonic increase in observed default rates, indicating affordability cut-offs could be refined for high-DTI segments.
- **Applicant Profile**: Loan purpose and geographic variation show meaningful differences in default behaviour, suggesting underwriting policy should account for borrower context and segment-specific risk.



Interactive Dashboard

Rank drivers by ODR variation to identify the most important risk-separating features.

Select a feature to inspect bucket-level default behaviour and segment concentration.

Use the dashboard link to explore risk patterns interactively across borrower segments.